Introduction to Probability

- Reading: Rosen 7.1, 7.2

Basic Definitions of Probability

- An experiment is a procedure that yields one of a given set of possible outcomes.
- The sample space *S* of the experiment is the set of all possible outcomes.
- An event *E* is a subset of the sample space.
- If S is a finite sample space of equally likely outcomes, and E is an event, i.e., a subset of S, then the probability of E is (Laplace definition of probability):

$$p(E) = \frac{|E|}{|S|}$$

Review

Experiment:
Roll a dice



Sample space: $S = \{1,2,3,4,5,6\}$

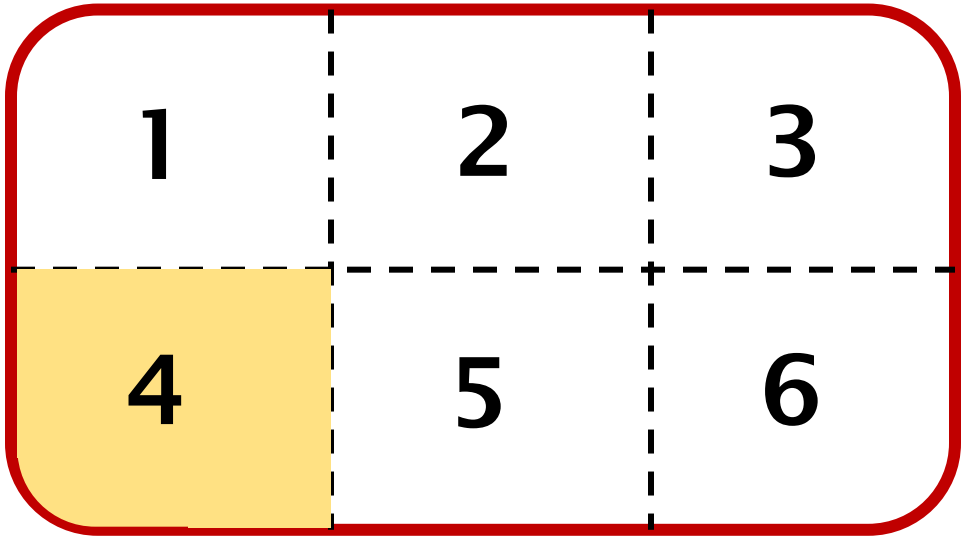
1	2	3
4	5	6

Review

Experiment:
Roll a dice



Sample space: $S = \{1,2,3,4,5,6\}$

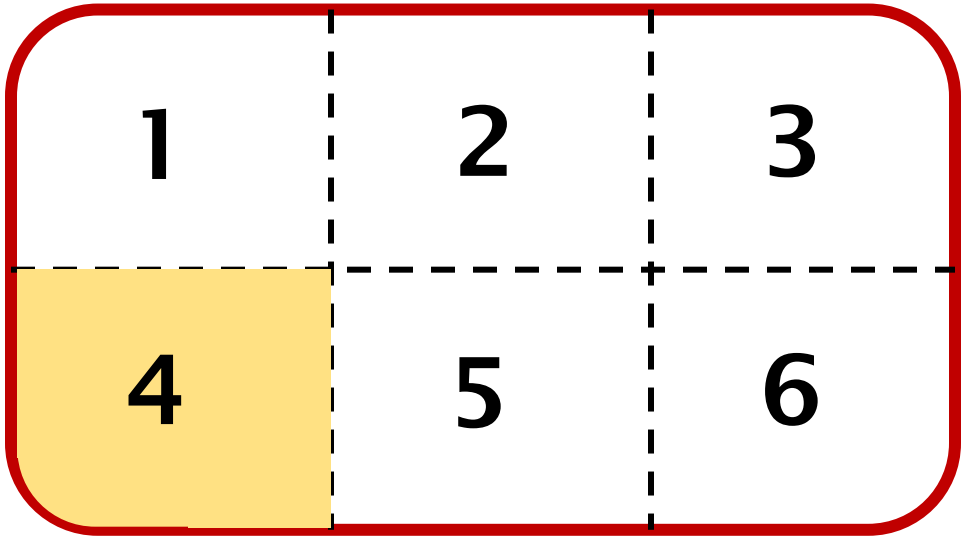


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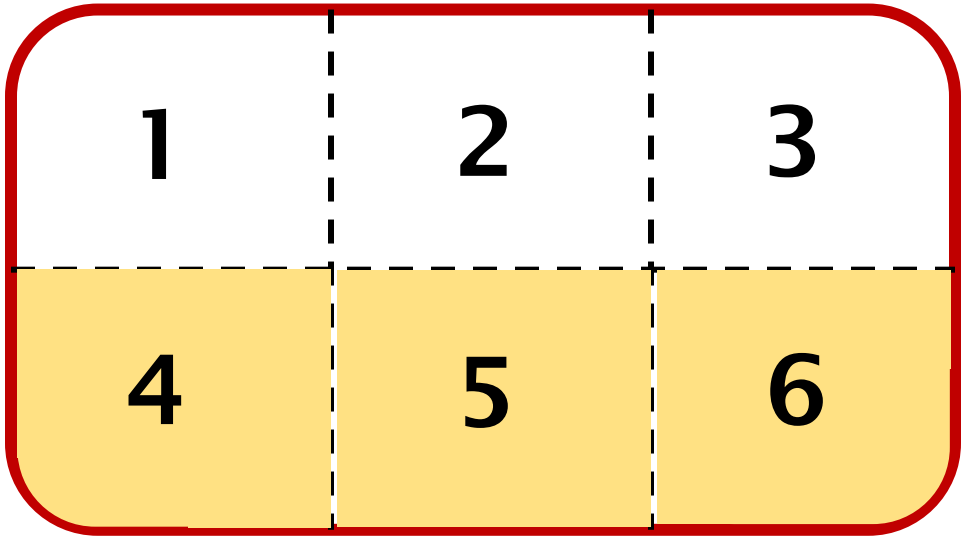
Every subset of S is an **event**
Event E = "outcome is 4"

Review

Experiment:
Roll a dice



Sample space: $S = \{1,2,3,4,5,6\}$



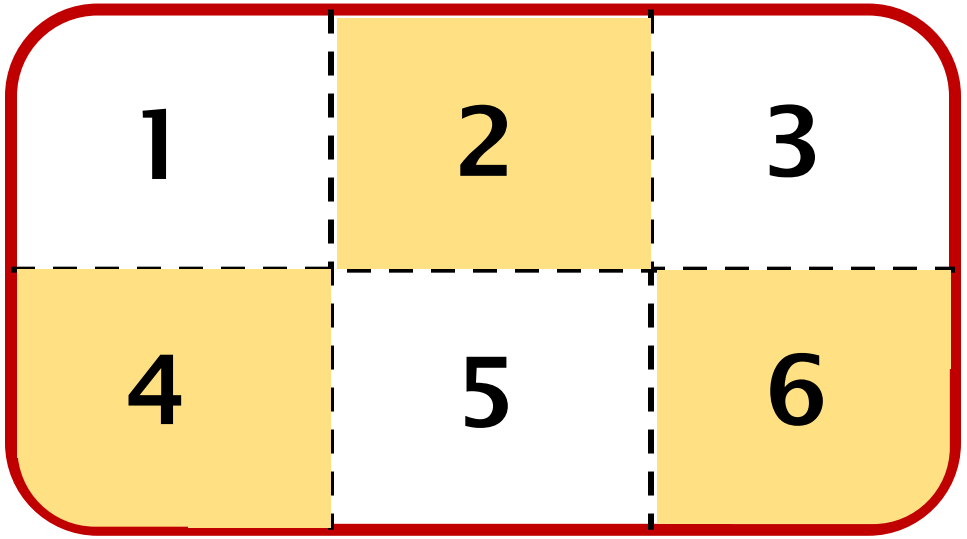
Every subset of S is an **event**
Event E = "outcome is 4"
Event F = "outcome is > 3"

Review

Experiment:
Roll a dice



Sample space: $S = \{1,2,3,4,5,6\}$



Every subset of S is an **event**
Event E = "outcome is 4"
Event F = "outcome is > 3"
Event G = "outcome is even"

Discrete Probability = Counting

- In Discrete Probability the set of outcomes is finite or countably infinite
- Example 1
- A bag contains 4 blue balls and 5 red balls. What is the probability that a ball chosen from the bag is blue?
- 4/9
- Example 2
- What is the probability that when two dice are rolled, the sum of the numbers on the two dice is 7?
- By the product rule there are $6^2 = 36$ possible outcomes. Six of these sum to 7. Hence, the probability of obtaining a 7 is $6/36 = 1/6$.

Exercise: Hatcheck Problem

- A new employee checks the hats of n people at a restaurant, forgetting to put claim check numbers on the hats. When customers return for their hats, the checker gives them back hats randomly. What is the probability that no one receives the correct hat?
- Answer: Number of derangements/Total number of permutations
- Total number of permutations: $n!$
- Number of derangements (from previous class):

$$D_n = n!\left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots (-1)^n \frac{1}{n!}\right]$$

- Probability of that a random permutation is a derangement:

$$p = \frac{D_n}{n!} = 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots (-1)^n \frac{1}{n!}$$

Lottery Exercises

- In a lottery, players win a prize when they guess a set of *m* numbers selected by a random process out of the first *n* positive integers (*n* is usually between 30 and 60, order is not important). What is the probability to win the prize?.
- There are C(*n*,*m*) combinations, and you have to guess the correct one.
- Probability of winning: 1/ C(*n*,*m*)
- In a different lottery, players win a large prize when they pick *n* (decimal) digits that match, *in the correct order*, *n* digits selected by a random process, where the same digit may be repeated.
- What is the probability that a player wins the large prize?
- The number of *n* permutations with repetition is 10^n. Therefore, the probability of guessing the correct permutation is: $1/\ 10^n$.
- A small prize is won if only *n*-1 digits are matched. What is the probability that a player wins the small prize?
- There are C(*n*,1)=*n* ways to select the incorrect digit.
- There are 9 choices for the value of the incorrect digit (all but the correct value).
- There is only 1 choice for the remaining *n*-1 correct digits.
- The probability that a player wins the small prize is: $9n\ /\ 10^n$ (winning outcomes/all outcomes).

Exercise on Letter Arrangements

- What is the probability of the following events when we randomly select a permutation of the 26 lowercase letters of the English alphabet?
- The first 13 letters of the permutation are in alphabetical order.
- There are 13! possible permutations of the first 13 letters and only 1 has them in alphabetical order. Thus, p=1/13!. Alternatively, since the first 13 letters are in order, we can only permute the remaining 13: $p=P(26,13)/26!=1/13!$
- The first letter of the permutation is A and the last letter is Z.
- Once I fix the first and the last letter, there are 24! permutations for the rest. Thus, p=24!/26!
- A and Z are next to each other in the permutation.
- I consider A and Z as a block. Thus, the number of possible permutations is 2x25! (2 because I need to select A or Z to be first). p= 2x25!/26!=1/13.
- A and Z are not next to each other in the permutation.
- p=12/13 (based on previous question).
- Z precedes A in the permutation.
- Z is equally likely to precede or follow A. Thus, p=1/2.
- Z precedes both A and B in the permutation.
- In 1/3 of permutations each of the 3 letters will be first. Thus, p=1/3.

Sampling

- What is the probability that the numbers 11, 4, 17, 39, and 23 are drawn in that order from a bin containing 50 balls labeled with the numbers 1, 2, …, 50 if
- The ball selected is not returned to the bin before the next ball is selected (sampling without replacement)?
- The number of 5 permutations of 50 balls is $P(50,5)$. The, probability of a single permutation is $\frac{1}{P(50,5)}$.
- The ball selected is returned to the bin before the next ball is selected (sampling with replacement)?
- The number of 5 permutations of 50 balls, with repetition, is 50^5. The, probability of a single permutation is $1/50^5$.

Complement of Event

- Theorem
  Let E be an event in a finite sample space S. The probability of the event $\bar{E}$, the complementary event of E, is given by $p(\bar{E}) = 1 - p(E)$.
- Proof
  Using the fact that $|\bar{E}| = |S| - |E|$,

$$p(\overline{E}) = \frac{|S|-|E|}{|S|} = 1 - \frac{|E|}{|S|} = 1 - p(E).$$

Example on Complement of Event

- A sequence of 10 bits is chosen randomly. What is the probability that at least one of these bits is 0?
- **Solution**:
  E: the event that at least one of the 10 bits is 0.
  $\overline{E}$: the event that all of the bits are 1s.
  The size of the sample space S is 2^{10}. Hence,

$$p(E) = 1 - p(\overline{E}) = 1 - \frac{|\overline{E}|}{|S|} = 1 - \frac{1}{2^{10}} = 1 - \frac{1}{1024} = \frac{1023}{1024}.$$

Poker Exercises on Event Complement

- A poker hand contains 5 out of 52 cards. The cards are 13 kinds (1-Ace, 2,..,10,J,Q,K) and 4 suites (spades, hearts, clubs, diamonds)
- What is the probability that a hand contains at least 1 Ace?
- I first compute the probability that the hand does not contain an Ace. In this case I have C(48,5) choices for the five cards. Therefore, the probability of no ace is: C(48,5)/ C(52,5).
- The probability of at least 1 ace is 1-C(48,5)/C(52,5).
- What is the probability that a hand does not contain the Ace of Spades:
- If we exclude the AoS, we can select all cards in C(51,5) ways.
- The probability of this is: C(51,5)/C(52,5)
- Alternatively, if we have the AoS, then we can select the remaining cards in C(51,4) ways. Thus, probability of having the AoS is: C(51,4)/C(52,5).
- The probability of not having the AoS is: 1- C(51,4)/C(52,5).
- C(51,5)/C(52,5) = 1- C(51,4)/C(52,5)

Exercise on Birthdays

- What is the probability of a person having today, assuming 366 days per year and uniform distribution of birthdays?
- 1/366
- Probability of a person not having birthday today:
- 365/366
- Probability that none of *n* persons has birthday today:
- $p(\text{no birthday today})=(365/366)^n$.
- Probability that in a group of *n* people, at least one has birthday today?
- $p(\text{at least one birthday today})=1-(365/366)^n$

The Birthday Problem

- **Question**
  What is the minimum number of people who need to be in a room so that there must be two people with the same birthday, assuming that there are 366 days in a year?
- **Answer**: 367 (pigeonhole principle)
- **Question**
  What is the minimum number of people who need to be in a room so that the probability that at least two of them have the same birthday is greater than 1/2, assuming that there are 366 days in a year, the birthdays of the people in the room are independent, and each birthday is equally likely?

The Birthday Problem: Solution

- Imagine the people entering the room one by one
- p_n: The prob. that the first n people all have different birthdays.
- $p_1 = 1$
- $p_2 = \frac{366 \times 365}{366^2} = 1 \times \frac{365}{366}$
- $p_3 = \frac{366 \times 365 \times 364}{366^3} = 1 \times \frac{365}{366} \times \frac{364}{366}$
- $p_n = 1 \times \frac{365}{366} \times \frac{364}{366} \times \cdots \times \frac{367-n}{366}$
- Want to find the maximum n such that $p_n < 1/2$. $1 - p_n > 1/2$ is the prob. that at least 2 among the n people have the same birthday.
- It turns out $p_{23} \approx 0.494$. Thus, for n=23 people there is $1 - p_{23} \approx 0.506$ probability that at least two people have the same birthday.

Union of Events

- **Theorem**
  Let E_1 and E_2 be events in the sample space S. Then

$$p(E_1 \cup E_2) = p(E_1) + p(E_2) - p(E_1 \cap E_2)$$

- **Proof**:
  Given the inclusion-exclusion formula $|A \cup B| = |A| + |B| - |A \cap B|$, it follows that

$$\begin{aligned} p(E_1 \cup E_2) &= \frac{|E_1 \cup E_2|}{|S|} = \frac{|E_1| + |E_2| - |E_1 \cap E_2|}{|S|} \\ &= \frac{|E_1|}{|S|} + \frac{|E_2|}{|S|} - \frac{|E_1 \cap E_2|}{|S|} \\ &= p(E_1) + p(E_2) - p(E_1 \cap E_2). \end{aligned}$$

Example of Union of Events

- What is the probability that a positive integer selected at random from the set of positive integers not exceeding 100 is divisible by either 2 or 5?
- **Solution**:

  E_1: the event that the integer is divisible by 2;
  E_2: the event that it is divisible 5;
  $E_1 \cup E_2$: The event that the integer is divisible by 2 or 5;
  $E_1 \cap E_2$: The event that it is divisible by 2 and 5.

  It follows that:

$$p(E_1 \cup E_2) = p(E_1) + p(E_2) - p(E_1 \cap E_2)$$
$$= 50/100 + 20/100 - 10/100 = 3/5.$$

Inclusion-Exclusion Principle for Probability

- **Theorem**
  Let $E_1, E_2, \ldots, E_n$ be events in a sample space S. Then

$$p\left(\bigcup_{i=1}^{n} E_i\right) = \sum_{1\le i\le n} p(E_i) - \sum_{1\le i<j\le n} p(E_i \cap E_j) + \sum_{1\le i<j<k\le n} p(E_i \cap E_j \cap E_k) - \cdots + (-1)^{n+1} p(E_1 \cap E_2 \cap \cdots \cap E_n).$$

- Disjoint union:
  If $E_1, E_2, \ldots$ is a sequence of pairwise disjoint events in a sample space S, then

$$p\left(\bigcup_i E_i\right) = \sum_i p(E_i)$$

Properties of Probability

- $p(S) = \frac{|S|}{|S|} = 1$
- $p(E) \leq 1$, for all events E
- $\bar{E}$ denotes the **complement** of E: $|\bar{E}| = |S| - |E|$
- $p(\bar{E}) = \frac{|S|-|E|}{|S|} = 1 - \frac{|E|}{|S|} = 1 - p(E)$
- $p(E \cup F) = \frac{|E \cup F|}{|S|} = \frac{|E|+|F|-|E \cap F|}{|S|} = p(E) + p(F) - p(E \cap F)$

Probability Distribution

What if outcomes are not equally likely?

Definition

Let S be the sample space of an experiment with a finite number of outcomes. A probability distribution on S is characterized by a probability mass function (pmf) $p: S \to \mathbf{R}$ where $p(s)$ is the probability of an outcome s, that is:

$p(s)$ *is the probability the outcome of the experiment is* s

It must hold that:

1. $0 \leq p(s) \leq 1$ for each $s \in S$
2. $\sum_{s \in S} p(s) = 1$

- The sample space maybe infinite.

Probability of an Event

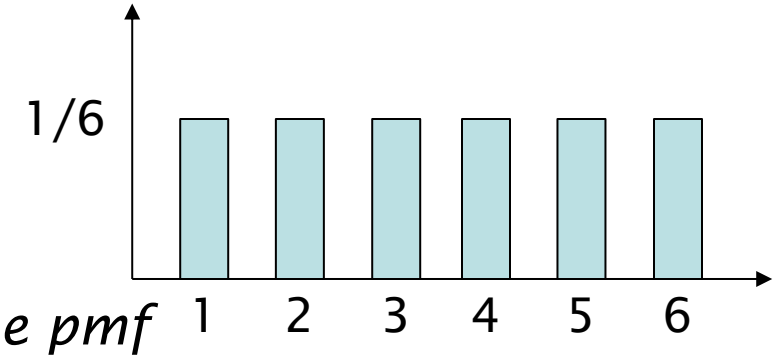
- **Definition**:
  The probability of the event E is the sum of the probabilities of the outcomes in E.

$$p(E) = \sum_{s \in E} p(s)$$

- Uniform distribution*: count*
- General distribution*: sum*
- $p(\bar{E}) = 1 - p(E)$ and Inclusion-Exclusion Principle are still true under the new definition.

Uniform Distribution

- **Definition**
  Suppose that S is a set with n elements. The uniform distribution assigns the probability $1/n$ to each element of S, i.e., all outcomes are equally likely.
- **Example**:
  For a fair dice with 6 sides, we have $p(s) = \frac{1}{6}$ for all s.

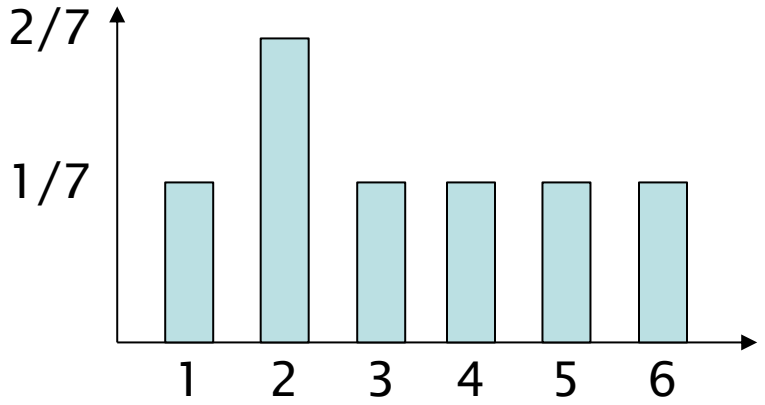


Histogram of the pmf

- Uniform distribution corresponds to the Laplace definition.

Non-Uniform Distribution

- **Example**:
  Suppose that a dice is biased so that
  (1) $p(2) = \frac{2}{7}$,
  (1) $p(s) = \frac{1}{7}$ for all other s.



Histogram of the pmf

- What is the probability that an even number appears when we roll this dice?

Independence

- **Definition**
  Two events E and F are **independent** if and only if
  $$p(E \cap F) = p(E)p(F)$$
- How to check independence?
  - Two unrelated events are independent; one does not affect the other.
  - Use definition
- Examples
- If two cards are drawn *with* replacement from a deck of cards, the event of drawing a red card on the first trial and that of drawing a red card on the second trial are *independent*.
- If two cards are drawn *without* replacement, the events are not independent because there are fewer red cards for the second trial.

Example

- In a randomly generated bit string of length 4:
  E: it begins with a 1
  F: it contains an even number of 1s.
  Are E and F independent?
- **Solution**
  - $p(E) = 1/2$.
  - $p(F) = 1/2$.
  - $p(E \cap F) = |\{1111, 1100, 1010, 1001\}|/16 = 1/4$.
  - $p(E \cap F) = p(E)p(F)$. So E and F are independent.

Exercise on Child Gender

- Each child has equal probability to be a boy or a girl
- *E*: the event that a family of *n* children has children of both sexes
- *F*: the event that a family of *n* children has at most one boy, respectively.
- Are *E* and *F* independent if *n*=2?
- $p(E) = \frac{|\{BG, GB\}|}{4} = \frac{1}{2}$
- $p(F) = \frac{|\{BG, GB, GG\}|}{4} = \frac{3}{4}$
- $p(E \cap F) = \frac{|\{BG, GB\}|}{4} = \frac{1}{2}$

- $p(E \cap F) = \frac{1}{2} \neq p(E)p(F) = \frac{3}{8}$. So they are not independent

Exercise 2 on Child Gender

- Suppose a family have 3 children, each of which has equal probability to be a boy or a girl. Are these two events independent?
  - E: The family has children of both sexes
  - F: The family has at most one boy
- **Solution**:
  - $p(E) = \frac{|\{BBG,BGB,BGG,GBB,GBG,GGB\}|}{8} = \frac{3}{4}$
  - $p(F) = \frac{|\{BGG,GBG,GGB,GGG\}|}{8} = \frac{1}{2}$
  - $p(E \cap F) = \frac{|\{BGG,GBG,GGB\}|}{8} = \frac{3}{8}$
  - $p(E \cap F) = p(E)p(F)$. So they are independent

Exercise 3 on Child Gender

- Each child has equal probability to be a boy or a girl
- *E*: the event that a family of *n* children has children of both sexes
- *F*: the event that a family of *n* children has at most one boy, respectively.
- Are *E* and *F* independent if *n*=5?
- View each child as a bit (e.g., 0 for boy, 1 for girl) in a bit-string of length 5: $p(E) = \frac{30}{32}$
- $p(F) = \frac{6}{32}$ (the bit-string has no 0, or it has a single 0)
- $p(E \cap F) = \frac{5}{32}$
- $p(E \cap F) = \frac{5}{32} \neq p(E)p(F) = \frac{45}{256}$. So they are not independent

Exercise 4 on Child Gender

- Each child has equal probability to be a boy or a girl
- *E*: the event that a family of *n* children has children of both sexes
- *F*: the event that a family of *n* children has at most one boy, respectively.
- Are *E* and *F* independent for any value of *n*>3?
- View each child as a bit (e.g., 0 for boy, 1 for girl) in a bit-string of length *n*: $p(E) = \frac{2^n - 2}{2^n}$
- $p(F) = \frac{n+1}{2^n}$
- $p(E \cap F) = \frac{n}{2^n}$
- For *E* and *F* to be independent it must hold:
- $\frac{n}{2^n} = \frac{2^n - 2}{2^n}\frac{n+1}{2^n} \Rightarrow n2^n = (2^n - 2)(n+1) \Rightarrow 2^n - 2n - 2 = 0$
- The equation is true only for *n*=3. For *n*>3, $2^n > 2n + 2$